output a description of the groups, just return a partitioning of the whole customers base, and then data scientists need to perform further analysis of the different morphotypes to understand how these groups are composed. Once this knowledge is derived, it can be used by marketing teams to tailor targeted campaigns towards each of these groups (or some of them, if some of the groups are crucial to the company's wealth): for instance, on a video streaming website (like Netflix), data scientists might be able to identify a group of users that will likely watch comedies, another group more interested in action movies, etc..

In real-world examples, customers have hundreds of features that are considered for marketing segmentation; here, for the sake of visualization and to make explanations easier, we will use a simplified example, with just two features: annual income and average monthly expenses on our e-commerce site. We will pick this example up again later in the chapter, but first, we will give you some more context and the tools to perform clustering analysis.

12.1 Intro to Clustering

In recent years, especially from the second half of the first decade of this century, a single branch of AI got so much momentum that now it's often considered, from media and public opinion, a synonymous of AI. I'm talking, of course, about *Machine Learning* (*ML*), which in turn has been lately (since ~2015) increasingly identified with *Deep Learning* (*DL*).